SS–C–646b
January 15, 1963

SUPERSEDING
Int. Fed. Spec. SS-C-00646a(GSA-FSS)
August 10, 1962 and
Fed. Spec. SS-C-646
December 10, 1943

FEDERAL SPECIFICATION

CRAYON, MARKING, LUMBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers extruded crayons for marking on green, wet, and dry lumber.

1.2 **Classification.**

1.2.1 *Type, compositions, and hardnesses.* Crayons shall be of but one type (see 1.1) and of the following compositions and hardnesses, as specified (see 6.1).

Composition I. — Graphite and clay base.

Composition II. — Clay base.

Hardnesses (see table I):

Very soft (graphite black only).

Soft.

Medium.

Hard.

1.2.2 *Sizes and color.* (See 3.4.)

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 — Boxes; Folding, Paperboard.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

PPP-B-665 — Boxes, Paperboard, Metal Stayed (Including Stay Material).

PPP-B-676 — Boxes, Set-Up, Paperboard.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

FSC 7510

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:13Z
Check the source to verify that this is the current version before use.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Composition.** The composition of the crayons shall be uniform throughout with respect to hardness, color intensity, and marking qualities. The crayons shall be extruded under pressure. The crayons shall be free from grit or other foreign substances which might impair their serviceability.

3.1.1 *Composition I.* Composition I crayons shall consist of a mixture of black graphite, a good grade of clay, and a satisfactory binder to produce smooth and uniform marking qualities.

3.1.2 *Composition II.* Composition II crayons shall consist of a mixture of pigment or other suitable coloring matter, a good grade of clay, and a satisfactory binder to produce the marking qualities specified in 3.1.1.

3.2 **Shape.** Crayons shall be hexagonal in shape and shall be provided with a point at one end. Points shall be formed by either making an angular cut across the axis of the crayon to form a 30° chisel point, or by tapering approximately 1/2 inch at one end of the crayon to a point not more than 1/4 inch in diameter. The other end of the crayon shall be approximately at right angles to the sides.

3.3 **Breaking strength.** Breaking strength of crayons shall be as indicated in table I for the respective hardnesses (see 4.4.1).

TABLE I.—*Sizes, breaking strength, and colors of crayons.*

Sizes		Breaking strength (min.)	Colors
Length (min.)	Width between opposite sides[1]		
Inches	*Inch*	*Pounds*	
4-1/2	1/2	Soft 15 Medium 20 Hard 25	Black Graphite black Blue Green Purple Red White Yellow
		Very soft 10	Graphite black
5	1	Hard 30	Red

[1] A minus tolerance of 1/32 inch will be permitted.

3.4 **Sizes and colors.** Crayons shall be of the sizes (lengths and widths) and colors shown in table I, as specified (see 6.1).

3.4.1 *Color fastness.* There shall be no appreciable fading of the color of the crayon when tested in accordance with 4.4.2.

3.5 **Bleeding.** There shall be no bleeding of the crayon markings when subjected to the test described in 4.4.3.

3.6 **Wrapper.** Each crayon shall be wrapped

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:13Z
Check the source to verify that this is the current version before use.

in a durable paper wrapper which shall be securely fastened to the crayon. The wrapper shall cover the sides of the crayon except that it shall be approximately 3/4 inch from the pointed end. Graphite crayons may be covered with an enameled coating in lieu of the paper wrapper.

3.7 **Identification marking.** Unless otherwise specified, crayons shall be marked with the name of the manufacturer, or trademark if of sufficiently known character to be readily identified, and marked or presented in accordance with standard commercial practice to clearly indicate the color of the crayon. Any additional standard commercial marking will be acceptable.

3.8 **Workmanship.** The workmanship shall be in accordance with standard commercial practice and crayons shall be free from breaks, cracks, chips, and any other defects which might affect the serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** Unless otherwise specified (6.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection and test of preparation for delivery requirements.* The preservation, packaging, packing, and marking for shipment of lumber crayons shall be subject to inspection and test by the Government inspector to determine compliance with the requirements of Section 5 of this specification. The contractor shall provide the necessary facilities and supplies for inspections and tests made prior to final acceptance.

4.2 **Lot.** A lot shall consist of all lumber crayons offered for delivery at one time.

4.3 **Sampling.**

4.3.1 *Sampling for examination.* Sample crayons shall be taken from each lot and examined for the presence of the defects listed in table II. The sample unit shall be one crayon. The sample size shall be in accordance with Standard MIL-STD-105, inspection level L-8. The acceptable quality level (AQL) shall be 4.0 percent defective; one class of defects.

TABLE II.—*Examination of crayons*

Examine	Defects
Coating.	Not wrapped or enameled as specified.
Color.	Not color specified.
Size.	Less than length specified. Width not as specified within tolerance.
Shape.	Not shape specified. Point not chiseled (as required). Point not tapered (as required).
Identification marking	Missing, illegible, incorrect.
Workmanship.	Wrapper insecurely fastened to crayon. Broken, cracked, chipped crayon. Hard particle or grit affecting marking ease.

4.3.2 *Sampling for tests.* Sample size for tests shall be in accordance with MIL-STD-105, inspection level L-4. The AQL shall be 4.0 percent defective.

4.3.3 *Sampling for defects in packaging, packing, and marking.* The sample unit for this examination shall be one shipping container. The lot size for this examination shall be expressed in units of shipping containers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:13Z
Check the source to verify that this is the current version before use.

The examination shall be in accordance with table III. The sample size for this examination shall be in accordance with MIL-STD-105, inspection level L-6. The AQL shall be 4.0 percent defective.

TABLE III.—*Examination of preparation for delivery*

Examine	Defects
Packaging.	Material and construction not as specified. Number in unit or intermediate packages not as indicated.
Packing.	Material and construction not as specified. Number in shipping container not as specified. Closures not accomplished by specified or required methods or materials.
Marking.	Illegible, incorrect, incomplete, omitted, or not as specified.

4.4 Tests. Tests shall be made using green, wet, and dry lumber.

4.4.1 *Breaking strength.* At 70° ± 4° F., six crayons of each composition, hardness, size, and color shall be tested for breaking strength (4.4.1.1). The crayon shall be supported in a horizontal position and equally at each end by two parallel rods 1/4 inch in diameter and 2-1/2 inches apart. Breaking strength load as specified in table I shall be applied gradually perpendicular to the axis of the crayon at the center and perpendicular to the plane of the support rods by means of a 1/4-inch-diameter rod placed parallel to the support rods, for a period of 5 minutes.

4.4.1.1 Paper-wrapped crayons shall be tested as specified above after removal of paper wrappers (3.6).

4.4.2 *Color fastness.* Six to eight lines shall be drawn with the sample crayon on one face of a piece of wood. Care shall be taken to maintain uniform pressure in drawing all lines. Approximately half of each of the lines shall be covered with suitable material to protect it from radiation from the arc lamp. The marked face of the wood shall be exposed for 40 hours to the radiation from a glass-enclosed carbon-arc lamp, such as the Fade-Ometer, with a current of approximately 13 amperes and 140 volts at the arc. The piece of wood shall be placed 10 inches from the arc. The lamp shall be so ventilated that the temperature at the face of the wood shall not exceed 105°F. (41°C.). There shall be no appreciable fading of the color of the exposed area as compared with the unexposed area.

4.4.3 *Bleeding.* Markings shall be made with the sample crayon on wood that shall then be immersed in water at 70° to 75° F. (21° to 24° C.) for a period of 24 hours. Examine for compliance with 3.5.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the various levels of packaging and packing specified hereinafter shall be as specified in Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A and B.*

5.1.1.1 *Unit packages.* Twelve crayons of one composition, hardness, color, and size shall be unit packaged in a set-up or folding paperboard box in accordance with the manufacturer's standard commercial practice.

5.1.1.2 *Intermediate packages.* Twelve unit boxes of crayons of one composition, hardness, color, and size, packaged as specified in 5.1.1.1, shall be packaged in an intermediate box. Intermediate boxes shall conform to PPP-B-676, type I or II, grade, class, and style optional; PPP-B-566, style, type, and class optional; PPP-B-636; or PPP-B-665, style optional.

5.1.2 *Level C.* Unless otherwise specified, unit and intermediate packaging shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:13Z
Check the source to verify that this is the currentversion before use.

5.1.2.1 *Unit packages.* Twelve crayons of one composition, hardness, color, and size shall be packaged in a box in accordance with standard commercial practice.

5.1.2.2 *Intermediate packages.* Twelve unit boxes of crayons, as specified in 5.1.2.1, shall be intermediate packaged in a box in accordance with standard commercial practice.

5.2 *Packing. Packing* shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Unless otherwise specified, level A packing shall be as follows: Twelve (12) intermediate boxes of crayons, as specified in 5.1.1.2, shall be packed in any one of the following shipping containers, at the option of the contractor:

Specification:	*Type and class*
PPP-B-585	Class 3 use
PPP-B-591	Overseas-type
PPP-B-601	Overseas-type
PPP-B-621	Class 2
PPP-B-636	Class 2

Each shipping container shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Case liners may be omitted for fiberboard boxes conforming to PPP-B-636 provided all corners, edge seams, and manufacturer's joints are sealed with minimum 1-1/2-inch-wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification, or appendix thereto, except that fiberboard boxes may be banded with filament-reinforced tape in accordance with the appendix to PPP-B-636.

5.2.2 *Level B.* Twelve intermediate boxes of crayons, as specified in 5.1.1.2, shall be packed in a fiberboard box conforming to PPP-B-636, type I or II, class 1, style RSC, bursting strength not less than 200 p.s.i. Closure of boxes shall be in accordance with the appendix to the specification.

5.2.3 *Level C.* Unless otherwise specified, crayons shall be packed in accordance with the manufacturer's standard commercial practice in containers conforming to carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order, marking shall conform to Fed. Std. No. 123 and shall be as specified hereinafter.

5.3.1.1 *Unit and intermediate packages.* Unit and intermediate packages shall be marked on one end or side with (1) applicable Federal Stock Number, (2) Federal Item Identification including specification classification (composition, hardness, color, and size), (3) unit and quantity (e.g. 1-DOZEN), or (e.g. 12-DOZEN) and name or trademark of the manufacturer.

5.3.1.2 *Shipping containers.* Shipping containers shall be marked in accordance with *Fed. Std.* No. 123, "Standard markings," as applicable, including specification classification such as composition, hardness, size, and color.

5.3.2 *Military agencies.* In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

5.4 Standard Pack (civil agencies). The Standard Pack for crayons shall be as specified in 5.1.1 (level B) and 5.2.2, including marking as specified in 5.3.1. (See 6.1.1.)

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein (5.1.2,

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:13Z
Check the source to verify that this is the current version before use.

5.2.1, and 5.2.3) and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Composition, hardness, size, and color (see 1.2.1 and 1.2.2).

(c) Inspection responsibility (*for civil agencies only*) if other than as specified herein; and administrative provisions for inspection records, when applicable (see 4.1).

(d) Level of packaging and packing and any special marking for shipment required (see 5.1, 5.2, 5.3.1, and 5.3.2).

6.1.1 *Standard Pack* requirements specified in 5.1.1 (level B) and 5.2.2, including marking as specified in 5.3.1, are intended for use in procurement of Stores stock replenishments. Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

6.2 Transportation data. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Crayons, marking, 1/4" or over in thickness.
Carload minimum weight 36,000 pounds.

Motor:

Crayons, marking.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, compositions, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:13Z
Check the source to verify that this is the current version before use.